

Royal Hospital
Department of Child Health

Management of complicated pneumonia in children guideline

First Edition: June 2023

Revision Date: June 2025

Background | Definitions:

- **Complicated pneumonia:** Chest infection associated with a complication such as parapneumonic effusion, empyema, lung abscess, or necrotizing pneumonia.
- **Parapneumonic effusion and empyema:**
 - Defined as pleural effusion associated with lung infection, mainly pneumonia.
 - Many parapneumonic effusions are associated with bacterial infections, most commonly by *S. pneumoniae*, *S. aureus* and *S. pyogenes*. They may also be caused by viral, fungal, mycoplasma pneumonias and tuberculosis.
 - Parapneumonic effusions evolve through three stages:
 - *Exudative:* sterile, free-flowing fluid, 2-5 days after the onset of the effusion.
 - *Fibro-purulent:* deposition of fibrin over the visceral and parietal pleurae, fluid becomes loculated or septated, 5-10 days after the onset of the effusion.
 - *Organized:* A thick and stiff pleural peel or rind develops and is attached to both visceral and parietal pleurae, 10-14 days after the onset of the effusion.
- **Necrotizing pneumonia:** Necrosis, and liquefaction of lung parenchyma
- **Lung abscess:** Accumulation of inflammatory cells, accompanied by tissue destruction or necrosis that produces one or more cavities in the lung.
- **Pneumatocele:** Thin-walled, air-containing cysts of the lungs

Scope of this guideline:

- **Inclusion Criteria:**
 - Diagnosis of pneumonia with a parapneumonic effusion or empyema, identified via chest radiography with suspected bacterial cause in children greater than 12 months to 12 years of age.
- **Exclusion Criteria: (Consult pulmonology or ID for any of the following)**
 - Children less than or equal to 12 months
 - Systemic illness concerning for sepsis
 - Cystic Fibrosis
 - Chronic lung disease
 - Immunodeficiency
 - Children on immunosuppressive medications
 - Sickle Cell Disease
 - Trauma

PICU/HD admission criteria

- Requirement of Oxygen therapy of more than 4L/min via nasal canula or facemask to maintain saturation $\geq 92\%$.
- Requirement of respiratory support (HFNC, NIV or Invasive ventilation) for increased work of breathing with/without respiratory failure (hypoxia or hypercapnia).
- If the patient needs inter-costal drainage insertion and intrapleural fibrinolysis.
- Post-surgical decortication or VATS (Video assisted thoracoscopic surgery).

Sub-speciality referrals:

- **Infectious disease:** Consult ID for complicated pneumonia requiring intensive care, with systemic illness concerning for sepsis or in case of lack of clinical improvement after 72 hours excluding fever which may last for several days.
- **Pulmonology:** Consult Pulmonology for those with necrotizing pneumonias, pneumatocele or lung abscess and in cases where clinical deterioration is seen despite appropriate management.

Laboratory investigations

- **CBC:** for Hb, WBC and differentials
- **C-reactive protein (CRP)**
 - As a baseline marker of inflammation
 - CRP may be used to help monitoring the response in patients who are not typically improving with initial therapy or those who present with systemic illness concerning for sepsis
- **Blood Culture**
- **Sputum sample/cough swap sample for culture**
- **Respiratory viral studies:** If available and clinically indicated (e.g., PICU or HD admission)
- **Urea and electrolytes**
 - To identify any features of Syndrome of inappropriate secretion of ADH (SIADH)
- Measuring albumin is not routinely recommended as it is a common finding in children with plural effusion and albumin infusion is seldom needed.

Imaging

Definitive diagnostic imaging should not be delayed as early treatment of parapneumonic effusion/empyema is associated with decreased morbidity, hospital costs, and length of stay

- **AP/PA chest X-ray (CXR)**
 - Should be performed in erect position, when possible, to estimate the approximate size of effusion which is an important factor that determines management.
 - A decubitus view is not necessary but can be considered
 - Follow-up chest radiographs is not recommended if the patient is clinically stable.
 - Repeat chest imaging in individuals who:
 - Do not exhibit improvement within 48-72 hours following initiation of appropriate antimicrobial therapy.
 - Post therapeutic intervention e.g., ICD insertion or VATS
- **Chest ultrasound (US):** Can be used to
 - Confirm the presence of a pleural fluid collection in unclear cases
 - Localize the effusion and determine the presence of septation
 - To guide the thoracentesis or drain placement
 - To differentiate pleural effusions from consolidated lung and peripheral lung abscess from empyema.
- **Chest computed tomography (CT)**
 - Should **NOT** be performed routinely because it does not provide any additional clinically useful information compared with that gained from ultrasonography and does not predict outcomes nor alter management.
 - Can be considered in the presence of diagnostic doubt (e.g., suspicion of malignancy or parenchymal lung disease) - **to be discussed with pulmonology team**

Antibiotics choice (see table below for doses)

If effusion > 1/3 of hemithorax:

- Non-toxic appearance: Ceftriaxone
 - In case of allergy to Ceftriaxone: Levofloxacin
- Second agent is required in the following case:
 - Add Vancomycin (Linezolid in case of allergy to Vancomycin)
 - Toxic/ill appearing child.
 - Septic shock.
 - Add Clindamycin (Linezolid or Vancomycin in case of known allergy or unavailability)
 - Known MRSA colonization.
 - Toxin mediated disease (e.g., toxic shock syndrome)
 - Necrotizing pneumonia
 - Multi-systemic infection
- The antibiotic regimen should be modified in accordance with the child's clinical response to therapy, blood/pleural fluid culture and other tests (e.g. 16s ribosomal test) results (if applicable), and in consultation with Pediatric Infectious Disease team.

Table: Antibiotics doses for complicated pneumonia in children

Antibiotic	Recommended Dose
Amoxicillin	90 mg/kg/day divided TID (30 mg/kg/dose)
Amoxicillin/Clavulanic acid	15 mg/kg/day TID of Amoxicillin/Clavulanic acid + 15 mg/kg/day TID of Amoxicillin
Ceftriaxone (IV)	50-100 mg/kg/day q24h (max: 2,000mg/day)
Clindamycin (IV or PO)	30-40 mg/kg/day divided TID (max PO: 1,800 mg/day, max IV: 2,700 mg/day)
Levofloxacin (IV or PO)	• Age 6 months to less than 5 years: 20 mg/kg/day divided q12h (max: 750mg/day) • Age 5 years and older: 10 mg/kg/day q24h (max: 750mg/day)
Vancomycin (IV)	40-60 mg/kg/day divided every q8h (maintain trough level 10-15)
Linezolid (IV or PO)	1 month to < 12 y: 10 mg/kg/dose IV/PO Q8H >= 12 y: 10 mg/kg/dose IV/PO Q12H

Corticosteroid

As there are no current evidence supporting the benefits of corticosteroids in complicated pneumonia, it should **NOT be** used routinely.

ICD insertion

- **Indications:**
 - Moderate-large effusion as per chest radiograph (CXR & US) and one of the following:
 - Toxic appearance
 - Respiratory distress/failure attributed to pressure effect from the effusion
 - Lack of clinical improvement in 48-72 hours
 - Complex effusion requiring fibrinolytic therapy
- **Relative Contraindications:**
 - Deranged coagulation
 - Pulmonary adhesions from previous surgery
 - Diaphragmatic hernia
- **ICD Vs Pigtail:**
 - Since there is no evidence that large bore chest drains confer any advantage, small drains (pigtail catheters) should be used whenever possible to minimise discomfort.
 - Use of pigtail catheters may reduce the incidence of tube displacement, need for postoperative opioid infusions and help in early mobilization
- **Drain Placement: (Consult pediatric surgery team or interventional radiology (IR) -if available- if outside PICU or if in HD after working hours/weekends)**
 - Small bore percutaneous drains (US guided pigtail) should be inserted at the optimum site suggested by chest ultrasound.
 - Large bore surgical drains should also be inserted at the optimum site suggested by ultrasound, but preferentially placed in the mid axillary line through the “safe triangle”.
 - Neither substantial force nor a trocar should ever be used to insert a drain.
 - A chest radiograph should be performed after insertion of a chest drain.
 - All chest tubes should be connected to a unidirectional flow drainage system (such as an underwater seal bottle) which must be kept below the level of the patient’s chest at all times.
 - A bubbling chest drain should never be clamped.
- **Labs to send from pleural fluid**
 - Gram stain and bacterial culture.
 - Differential cell count.
 - Molecular testing (streptococcus pneumonia PCR) or broad range PCR techniques if available (16s rRNA)
 - If pleural lymphocytosis present, Tuberculosis and malignancy must be excluded by sending the sample for cytology, stain for Acid Fast Bacilli, mycobacteria culture and mycobacteria Tuberculosis PCR.
 - Biochemical analysis of pleural fluid is NOT necessary in the management of uncomplicated parapneumonic effusions/ empyema in children.
 - If there is any indication the effusion is not secondary to infection, consider an initial small volume diagnostic tap for cytological analysis, avoiding general anaesthesia/sedation whenever possible.

Thrombolysis

- **Indications:**
 - Every time an ICD is inserted, intra-plural fibrinolysis should be administered
- **Contraindications:**
 - Known hypersensitivity to fibrinolytics
 - Active pleural bleeding
 - Major surgery in the previous 5 days
 - Significant coagulopathy
 - Bronchopleural fistula
 - Necrotizing Pneumonia (relative contraindication, to discuss with Pulmonology)
- **Agent for thrombolysis:**
 - No evidence that any one of the 3 fibrinolytics (e.g., streptokinase, urokinase, or alteplase) are more effective than another.
 - Alteplase: (once daily for 3 days)
 - 0.1 mg/kg diluted in 10 to 100 mL of normal saline. Maximum dose is 6 mg
 - Or based on weight as following:
 - ≤ 10kg: 1mg in 20 ml NS
 - >10kg to 20kg: 2mg in 40 ml NS
 - >20kg to 30kg: 3mg in 40 ml NS
 - >30kg: 4mg in 40 ml NS
 - Streptokinase: (Once daily for 3 days)
 - 15,000 IU/kg diluted in 50ml normal saline (maximum dose 250,000 units)
 - Urokinase: (twice daily for 3 days)
 - < 10 kg: 10,000 units in 10 ml of NS
 - > 10 kg: 40,000 units in 40 ml of NS
- **How to instill intrapleural fibrinolytic agents?**
 - To be done under aseptic precautions
 - Be sure that chest drain in place and no contraindication in present
 - Explain to the patient/family about possible complications
 - Steps:
 - After preparing the medication (see the doses above), put it in a catheter tip syringe (The syringe with cone shaped tip)
 - Clamp the chest tube proximal to the connection between the chest tube and underwater seal bottle
 - Detach the chest tube from underwater seal bottle connector
 - Attach the syringe to the chest tube and instill the medication
 - Reattach the chest tube to the underwater seal bottle
 - Keep the chest tube clamped for 4 hours
 - Negative pressure is not typically required

Patient monitoring while ICD in place

- Vitals signs (HR, RR, BP, SPO2) every 15 minutes in the initial hour post insertion, then every 1 hour for 4 hours
- Pain score
- Accurate measurement of chest tube drainage (every 1 to 4 hourly):
 - ICD need be clamped for 1 hour if 10 ml/kg of fluid was drained at insertion or any time the drain amount is ≥ 10 ml/kg/hr
- Bleeding at the site of chest tube insertion or hemorrhagic plural drain
- The position and the level of external chest tube to avoid dislodgement

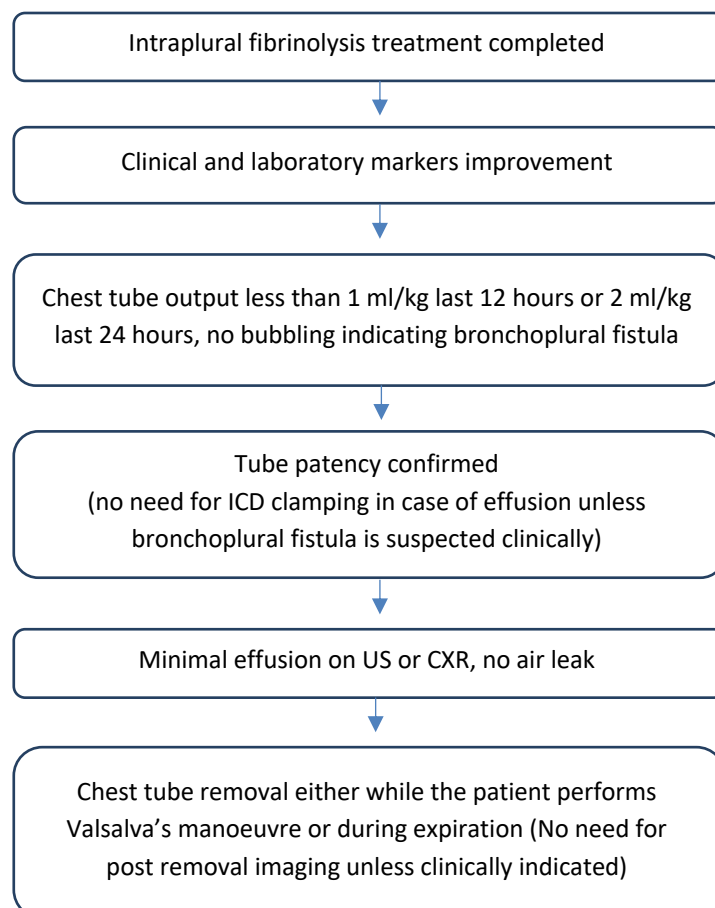
Pain management

- It is vital to ensure that the child receives adequate analgesia. They should be comfortable enough to mobilize and to take a deep breath.
- Simple regular analgesics such as paracetamol (10-15 mg/kg/dose every 6 hours) should be sufficient in most children who have a chest drain in situ.
- Tramadol 1-2 mg/kg/dose every 6-8 hours can be considered if significant pain noticed despite regular paracetamol.

Physiotherapy

- There is no evidence to support the use of chest physiotherapy in children with CAP
- Early mobilization is recommended

ICD removal



Video assisted thoracoscopy (VATS)

- Primary VATS as an initial treatment is rarely needed and required only in cases where the empyema is in the late organised stage (after 14 days of the development of effusion)
- VATS can be considered if the initial management with ICD and fibrinolysis failed which is defined as the persistence of fever at day 4 post completion of thrombolysis with demonstration of persistent loculation in the plural cavity without an intraparenchymal explanation of fever i.e., development of necrotizing pneumonia or abscess.
- In these selected cases, the need to VATS to be discussed between pediatric pulmonology and pediatric surgery teams for a shared decision making

Discharge criteria

- Removal of chest tube
- Off supplemental oxygen for 24 hours (acceptable saturation >90%)
- Able to tolerate oral intake including oral antibiotics trial in hospital

Oral antibiotics

- Depending on child's response to therapy and microbiological investigations results.
- Oral antibiotic options may include high dose Amoxicillin (90 mg/kg/day), Amoxclav, Cefuroxime or Clindamycin (in case of MSSA or MRSA).
- In most children, treatment for a total course of 3-4 weeks is adequate

Follow up plan

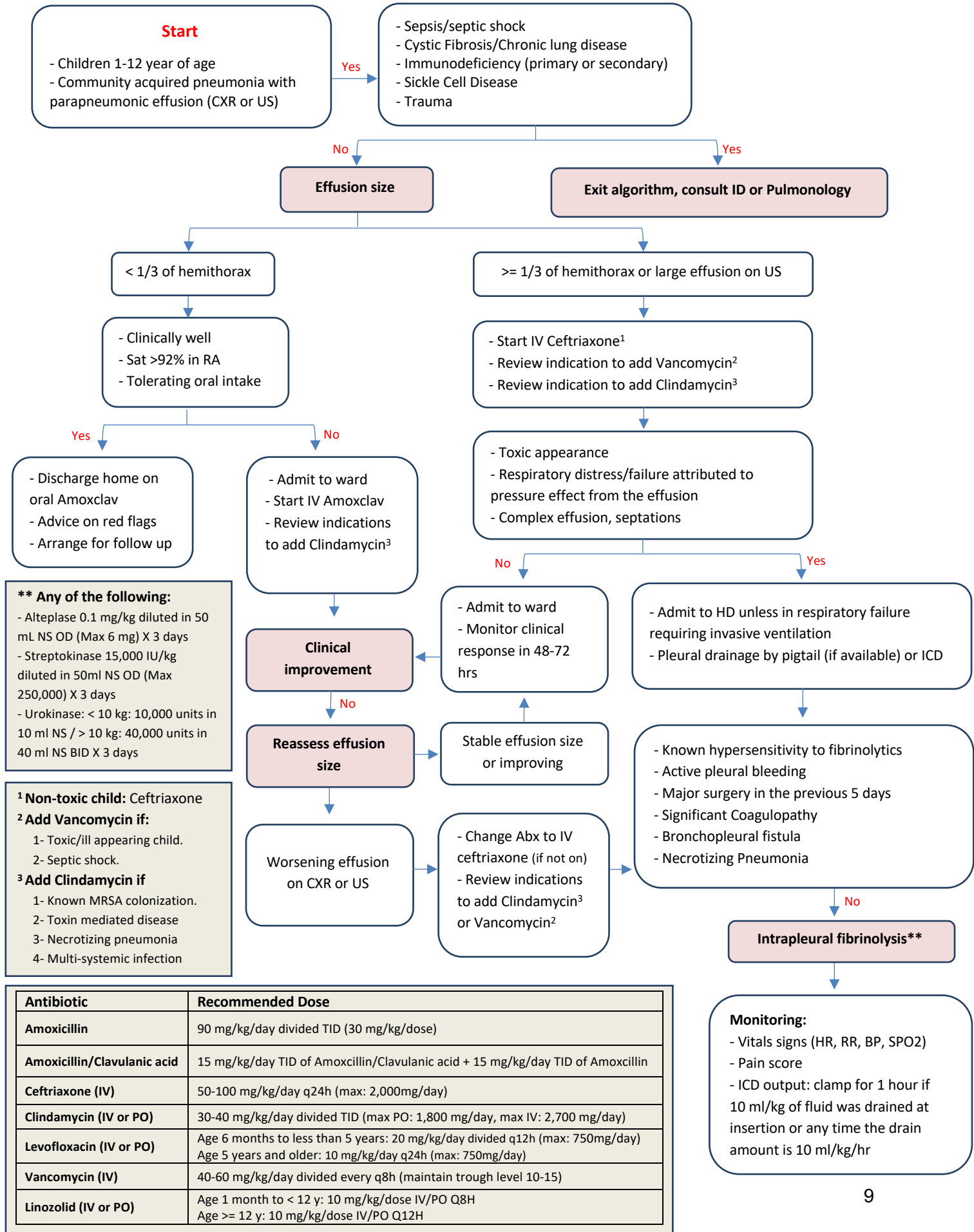
- General pediatrics OPD follow up within 4–6 weeks of discharge, the timing depending on the child's clinical status at discharge.
- Pulmonology follow up in cases of bronchopleural fistula or lack of improvement at 6 weeks follow up appointment.
- ID follow up in cases of multi-systemic infections

Additional information

Pneumococcal antibiotics susceptibility in 2020 (*Source: Central public health lab, MOH*)

- Penicillin susceptibility (meningitis): 60%
- Ceftriaxone susceptibility (meningitis): 98.2%
- Penicillin & Ceftriaxone susceptibility (non- meningitis): 100%
- Meropenem susceptibility: 94.6%
- Vancomycin: 100%

Algorithm: Management of complicated pneumonia with parapneumonic effusion



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